

Tim Jacobs

Senior/Staff Software Engineer | Data, AI & Platform Architecture

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[in](#) Tim Jacobs

SUMMARY

Staff Software Engineer with deep expertise in large-scale data platforms, LLM-powered retrieval systems (RAG), and cloud-native infrastructure. Proven architect of event-driven pipelines, vector search applications, and shared platform components that power data-intensive products. 15+ years building high-leverage backend systems from storage validation to distributed AI services. Seeking to apply this experience to mission-critical domains where technical excellence drives positive human impact.

SKILLS

Data Engineering, AI & Platform Architecture

Expert

Snowflake dbt Apache Airflow
RAG architecture
Vector search & embeddings
Event-driven systems

AI & Machine Learning

Accomplished

Snowflake Cortex
LLM integration & prompting
Streamlit
Custom tokenization & vectorization
k-NN / semantic search

Cloud Architecture & DevOps

Expert

AWS Kubernetes Terraform
Docker CI/CD pipeline design
Microservices architecture

Software Development

Expert

Python SQL Go/Golang
REST API FastAPI

Databases & Storage

Expert

PostgreSQL Redis
DynamoDB Elasticsearch
Database design & optimization

Engineering Leadership

Expert

Technical architecture & roadmap
Team mentorship & growth

WORK EXPERIENCE (4)

Dec 2022 - Current

Senior Software Engineer at [EDO Inc.](#)

Data platform & AI systems architecture for cross-platform content intelligence.

- Architected and deployed an internal RAG-based data-quality tool using Snowflake Cortex for LLM inference, custom tokenization and vectorization pipelines, and k-NN/embedding similarity search to surface AI-generated taxonomy mappings.
- Built a Streamlit review interface so the data quality team can inspect, validate, and accept AI recommendations produced by the RAG pipeline.
- Implemented medallion architecture on Snowflake-based data pipelines to ingest and process trillions of rows of cross-platform data, supporting 20+ enterprise partners.
- Engineered a hypernormalized metadata management system that standardized fragmented data from dozens of sources into a single canonical schema, achieving 80% coverage of all monthly events and eliminating manual mapping

bottlenecks.

Oct 2020 - Dec 2022

Staff Software Engineer at [Pie Insurance](#)

Full-stack platform engineering, quality-focused technical leadership, and operational data architecture.

- Architected and deployed 40+ shared platform components consumed by multiple engineering teams; reduced time-to-market for new features by 50% and improved code reuse across the organization.
- Led technical design of CI/CD pipeline modernization; improved deployment velocity by 3x, reduced mean time to recovery by 60%, and enabled 200+ deployments per week.
- Designed comprehensive test architecture for an event-driven billing service; enabled continuous deployment across 8+ microservices-based systems with 99.9% reliability.
- Led operational data team; designed and implemented reference data re-architecture using event-driven ETL; reduced mean commit-to-deploy time from weeks to hours, increasing release velocity by 3x.
- Developed scalable test automation framework for microservices in Python/Golang using AWS Lambda, S3, and SQS; automated 90% of previously manual test execution, reducing test cycle time from 6 hours to 12 minutes.
- Mentored 6 junior engineers on microservices patterns, testing strategy, and AWS cloud best practices; 3 engineers promoted during tenure.

Apr 2016 - Oct 2020

Senior Software Engineer at [NetApp - Solidfire](#)

Platform architecture, DevOps transformation, and test infrastructure leadership.

- Designed and led a test result reporting platform from concept to production; engineered a queryable database with 2M+ daily test records and automated management dashboards, eliminating 40+ hours per week of manual reporting.
- Spearheaded a DevOps transformation initiative with cross-functional stakeholders; reduced build failure rate by 50%, improved deployment reliability by 35%, and decreased infrastructure costs by \$500K+ per year.
- Architected a containerized test framework enabling migration of 1,000+ test cases from legacy systems to modern CI-driven infrastructure; reduced test execution time by 70% and improved reproducibility.

Jun 2013 - Apr 2016

SSD Validation Engineer → **Engineering Supervisor** at [Micron Technology](#)

Validation leadership, data-driven operations, and team management (4 direct reports).

- Managed a team of 4 engineers validating enterprise/data center SAS/SATA SSDs across 4 Tier-1 OEM partners; maintained >\$60M quarterly revenue through rigorous, on-time qualification processes.
- Designed and deployed a qualification test status database with automated test script integration; eliminated manual reporting, enabled real-time visibility into validation progress, and reduced management reporting overhead by 30 hours per week.
- Led a complete automation rewrite to meet compatibility team requirements; reduced deployment schedule by 20 months, improving test execution efficiency by orders of magnitude.

EDUCATION (1)

- 2005

Bachelor of Science Electrical Engineering at University of Colorado at Boulder

Grade: 3.74 (Cum Laude)